PES UNIVERSITY (ECC) DEPARTMENT OF CSE

SOFTWARE ENGINEERING

LAB 1 | Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT #60

Podcast Guest Scheduling & Outline Builder

Media, Events & Community

SUBMISSION BY	SRN	SECTION
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COMBINED LAB SUBMISSION

Problem Statement • Requirements Table • UML Use-Case Diagram • Use-Case Flow Specification

Problem Statement

1. Problem Context & Overview

A media-production scheduling tool that allows podcast guests to share booking availability, lets show hosts select suitable recording slots, collects structured guest talking-point outlines, and attaches the approved outline to the scheduled episode for preparation and production.

2. Target Stakeholders / Actors

Podcast Guest **Show Host**

Sample Functional Requirement: FR-001 [Priority: High]
Description: The system shall allow a podcast guest to publish available recording time slots and submit a structured outline of talking points for the selected episode.
Sample Acceptance Criteria: Pass: the selected slot is available and the outline is saved against the episode; Fail: overlapping bookings or an empty required outline are accepted.

3. Sample Non-Functional Requirement (NFR) Guideline
Sample Non-Functional Requirement: NFR-001 [Type: Performance & Security]
Description: The system shall generate the finalized episode schedule and attached guest outline within 2 seconds for normal workloads while preventing unauthorized users from viewing or modifying another user's booking information.
Sample Acceptance Criteria: Pass: performance tests meet the target latency and unauthorized access attempts are denied; Fail: scheduling requests exceed the target or expose protected data.

4. Required Deliverables

>> **Complete Requirements Table** — exactly 5 Functional Requirements (FR-001 to FR-005).

>> **2 Non-Functional Requirements** (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

>> **UML Use-Case Diagram** — model all actors and primary use cases and include at least one «include» and one «extend» relationship.

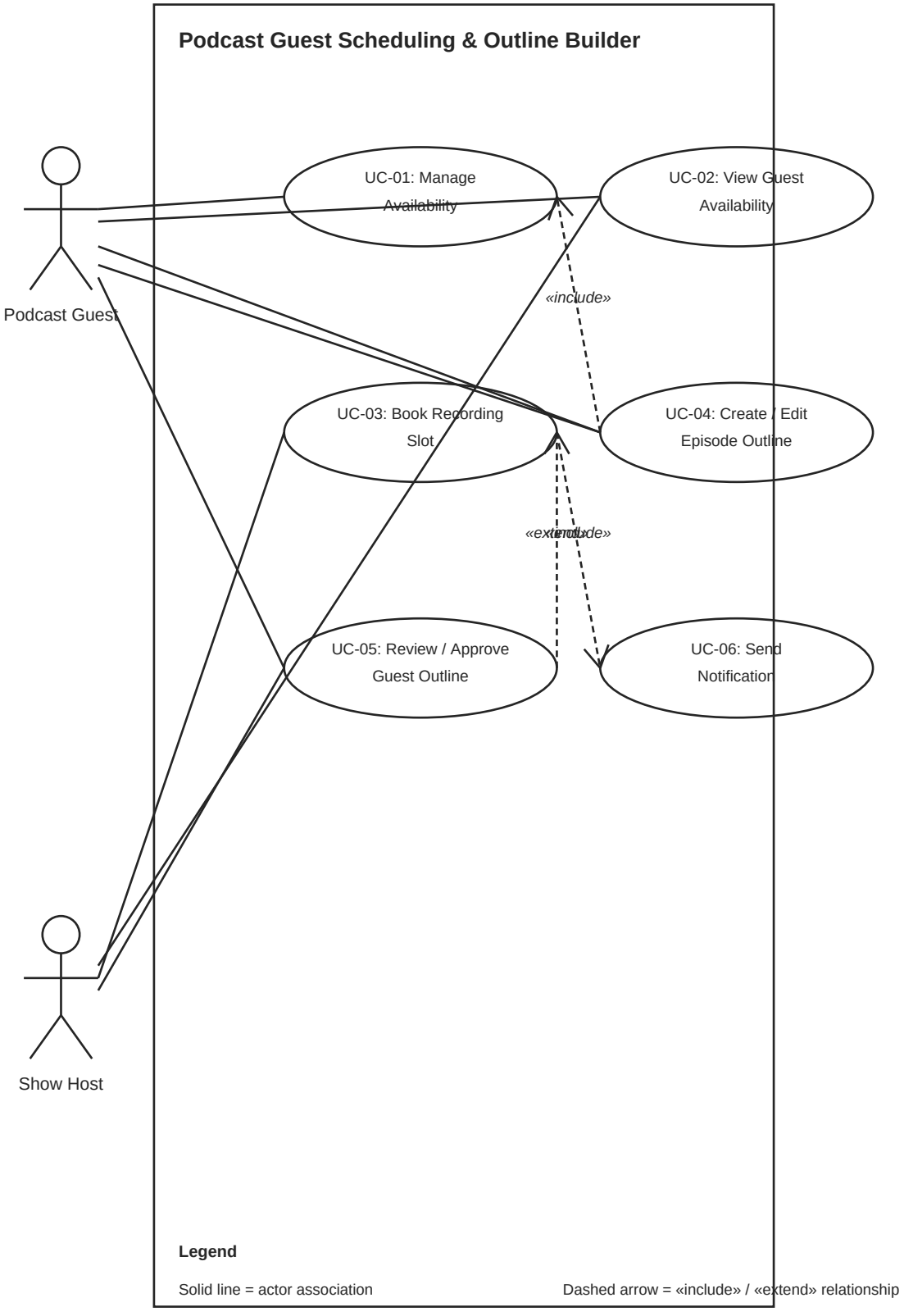
>> **Use-Case Flow Specification** — 1-page document for one core use case with Preconditions, Postconditions, Main Success Scenario, and one Alternate Flow.

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall allow a Podcast Guest to publish, update, and withdraw available recording time slots.	High	Only valid future time slots are saved; overlapping unavailable slots are rejected.	Guests need control over their booking availability before a host selects a time.	Availability must be checked before a slot is offered to a host.
FR-002	Functional	The system shall allow a Show Host to view a guest's available slots and book one suitable recording time.	High	A booking is created only when the selected slot is still available; the booked slot becomes unavailable.	The core scheduling function connects guest availability with host booking.	Booking confirmation should include episode and scheduled time.
FR-003	Functional	The system shall allow a Podcast Guest to create and attach structured talking-point outlines to a scheduled episode.	High	Required outline fields are validated and the saved outline is linked to the correct episode.	The outline is needed for guest preparation and episode planning.	Outline should support ordered bullet points and optional topic references.
FR-004	Functional	The system shall allow a Show Host to review the submitted guest outline and mark it as approved for the episode.	High	Only the assigned host can approve; approval status and timestamp are stored.	Host review provides a clear handoff from guest preparation to production.	Approval should preserve the latest submitted outline version.
FR-005	Functional	The system shall notify the Podcast Guest and Show Host when a booking is confirmed, changed, cancelled, or an outline is approved.	Medium	Notifications are generated for each supported event and reference the correct episode.	Timely notifications reduce missed recordings and coordination errors.	Notification content should state the event, episode, date/time, and action required.
NFR-001	Nonfunctional – Performance & Security	The system shall return normal availability and scheduling operations within 2 seconds and protect booking and outline data from unauthorized access.	High	Performance tests meet the 2-second target under normal load and unauthorized requests are denied.	Scheduling is time-sensitive and guest information should only be visible to authorized users.	Authentication and authorization must cover guest and host roles.
NFR-002	Nonfunctional – Reliability	The system shall preserve confirmed bookings and approved outlines consistently even if a notification delivery attempt fails.	High	A simulated notification failure does not remove or duplicate the stored booking or outline state.	The schedule and production outline are the source of truth and must not depend on notification delivery.	Failed notifications should be retryable without changing the saved episode state.

UML Use-Case Diagram

Actors, primary use cases, associations, and «include» / «extend» relationships



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Use-Case Flow Specification

UC-03 — Book Recording Slot

Primary Actor	Show Host
Supporting Actors	Podcast Guest
Preconditions	Show Host is authenticated; the guest has published at least one future availability slot; the episode exists and is ready for scheduling.
Postconditions	A confirmed recording booking is stored for the episode and selected guest slot; the selected slot is no longer available for another booking; both actors are notified.

Main Success Scenario

1. Show Host selects the Book Recording Slot function for the required episode.

2. System retrieves the selected Podcast Guest's currently available future time slots.

3. System displays the available slots with the relevant date and time information.

4. Show Host selects a suitable recording slot.

5. System validates that the selected slot is still available and is not already booked.

6. System creates the booking and associates the guest, host, episode, and selected time slot.

7. System marks the selected availability slot as booked so it cannot be selected again.

8. System stores the booking confirmation timestamp and current booking status.

9. System sends a confirmation notification to the Podcast Guest and Show Host.

10. System displays the confirmed recording time and booking status to the Show Host.

Alternate Flow — Selected Slot Becomes Unavailable

1. System detects that the selected availability slot has been booked or withdrawn after the host opened the availability list.

2. System does not create the booking and does not mark the episode as confirmed.

3. System informs the Show Host that the slot is no longer available and refreshes the available slots.

Use-Case Flow Specification

UC-03 — Book Recording Slot (continued)

Alternate Flow — Selected Slot Becomes Unavailable (continued)

4. Show Host selects another available slot.

5. System repeats the availability validation before creating the booking.

6. After successful validation, the system completes the normal booking flow.

Traceability Note
This flow operationalizes FR-002 and FR-005 by showing how a Show Host selects an available guest slot, creates a confirmed recording booking, prevents double booking, and triggers scheduling notifications. It also supports NFR-001 by requiring authorized access and timely scheduling operations.

Relationship Notes

• **«include»:** Booking a recording slot includes validating the selected availability before confirmation.

• **«include»:** Confirming a booking includes sending the corresponding scheduling notification.

• **«extend»:** Reviewing/approving the guest outline can extend the episode preparation flow after an outline has been submitted.